MAT 2040 Linear Algebra

Assignment 8

Please note that

- Released date:
- Due:
- Late submission is NOT acceptable.
- Please submit your assignment as a PDF file titled "student ID + HW8".

Question 1. Which of the following sets of vectors form an orthonormal basis for $\mathbb{R}^2$?

(a) $\left\{(1,0)^T,(0,1)^T\right\}$

(b) $\left\{\left(\frac{3}{5},\frac{4}{5}\right)^T,\left(\frac{5}{13},\frac{12}{13}\right)^T\right\}$

(c) $\left\{(1,-1)^T,(1,1)^T\right\}$

(d) $\left\{\left(\frac{\sqrt{3}}{2},\frac{1}{2}\right)^T,\left(-\frac{1}{2},\frac{\sqrt{3}}{2}\right)^T\right\}$

Solution

(a) Take $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ The inner product of these vectors is calculated as follows:

$$\begin{aligned} \langle \mathbf{v}_1, \mathbf{v}_2 \rangle &= \mathbf{v}_2^T \mathbf{v}_1 \\ &= \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= 0(1) + 1(0) \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

Therefore, the inner product of the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0$.

Find the inner product with the same vectors.

$$\langle \mathbf{v}_1, \mathbf{v}_1 \rangle = \mathbf{v}_1^T \mathbf{v}_1 = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1(1) + 0(0) = 1 + 0 = 1 \, \langle \mathbf{v}_2, \mathbf{v}_2 \rangle = \mathbf{v}_2^T \mathbf{v}_2$$

$$= \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0(0) + 1(1) = 0 + 1 = 1$$

Since the inner products $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0, \langle \mathbf{v}_1, \mathbf{v}_1 \rangle = 1, \langle \mathbf{v}_2, \mathbf{v}_2 \rangle = 1$, so by the definition of an orthonormal set, $\left\{(1,0)^T, (0,1)^T\right\}$ is an orthonormal. And, no vector of the set $\left\{(1,0)^T, (0,1)^T\right\}$ is a scalar multiple of the other, so it is a linearly independent set and hence it is an orthonormal basis in $\mathbb{R}^2$.
(b)The inner product of these vectors is calculated as follows:

$$\begin{aligned} &\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = {\mathbf{v}_2}^T \mathbf{v}_1 \\ &= \left(\frac{5}{13} \frac{12}{13} \right) \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix} \\ &= \frac{5}{13} \left(\frac{3}{5} \right) + \frac{12}{13} \left(\frac{4}{5} \right) \\ &= \frac{3}{13} + \frac{48}{65} \\ &= \frac{15 + 48}{65} \\ &= \frac{63}{65} \\ &\neq 0 \end{aligned}$$

Since the inner product of $\mathbf{v}_1 = \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 5/13 \\ 12/13 \end{pmatrix}$ is not equal to 0 , that is $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle \neq 0$. Therefore, by the definition of an orthonormal set, the set $\left\{ \left(\frac{3}{5}, \frac{4}{5}\right)^T, \left(\frac{5}{13}, \frac{12}{13}\right)^T \right\}$ is not an orthonormal. Hence, the set $\left\{ \left(\frac{3}{5}, \frac{4}{5}\right)^T, \left(\frac{5}{13}, \frac{12}{13}\right)^T \right\}$ is not an orthonormal basis for $\mathbb{R}^2$.
(c) The inner product is 0 , and the the inner product with the same vectors is caculcated as follows: $\langle \mathbf{v}_1, \mathbf{v}_1 \rangle = \mathbf{v}_1^T \mathbf{v}_1 = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1(1) +$ $(-1)(-1) = 1 + 1 = 2 \neq 1$ Since the inner product $\langle \mathbf{v}_1, \mathbf{v}_1 \rangle \neq 1$, so by the definition of an orthonormal set, the set $\left\{(1,0)^T, (0,1)^T\right\}$ is not an orthonormal. Hence, the set $\left\{(1,-1)^T, (1,1)^T\right\}$ is not an orthonormal basis for $\mathbb{R}^2$.
(d) Similarly, we can get that this vector set is an orthonormal basis in $\mathbb{R}^2$.

Question 2. Let

$$\mathbf{u}_1 = \begin{pmatrix} \frac{1}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$$

(a) Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis for $\mathbb{R}^3$.

(b) Let $\mathbf{x} = (1, 1, 1)^T$. Write $\mathbf{x}$ as a linear combination of $\mathbf{u}_1, \mathbf{u}_2$, and $\mathbf{u}_3$ and compute $\|\mathrm{x}\|$.

Solution

(a) We show $\{u_1, u_2, u_3\}$ is an orthonormal basis for R^3, for this we show $u_1^T u_1 = 1, u_1^T u_2 = 1$

$$u_3^T u_3 = 1, u_1^T u_2 = 0, u_1^T u_3 = 0, u_2^T u_3 = 0$$

Consider

$$\begin{aligned} u_1^T u_1 &= \frac{1}{18} + \frac{1}{18} + \frac{16}{18} = 1 \\ u_2^T u_2 &= \frac{4}{9} + \frac{4}{9} + \frac{1}{9} = 1 \\ u_3^T u_2 &= \frac{1}{2} + \frac{1}{2} + 0 = 1 \\ u_1^T u_2 &= \frac{\sqrt{2}}{9} + \frac{\sqrt{2}}{9} - \frac{2\sqrt{2}}{9} = 0 \\ u_1^T u_3 &= \frac{1}{6} - \frac{1}{6} + 0 = 0 \\ u_2^T u_3 &= \frac{\sqrt{2}}{3} - \frac{\sqrt{2}}{3} + 0 = 0 \end{aligned}$$

The set $\{u_1, u_2, u_3\}$ forms an orthogonal basis for R^3.

(b) Let $x = (1, 1, 1)^T x$ as the linear combination of u_1, u_2 and u_3 is given by $x = c_1 u_1 + c_2 u_2 + c_3 u_3$, where

$$c_1 = x^T u_1 = (1, 1, 1) \begin{bmatrix} \frac{1}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} \end{bmatrix} = \frac{1}{3\sqrt{2}} + \frac{1}{3\sqrt{2}} - \frac{4}{3\sqrt{2}} = \frac{-2}{3\sqrt{2}} = \frac{-\sqrt{2}}{3}$$

$$c_2 = x^T u_2 = (1, 1, 1) \begin{bmatrix} 2/3 \\ 2/3 \\ 1/3 \end{bmatrix} = \frac{2}{3} + \frac{2}{3} + \frac{1}{3} = \frac{5}{3}$$

$$c_3 = x^T u_3 = (1, 1, 1) \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 0 = 0$$

Therefore $x = -\frac{\sqrt{2}}{3}u_1 + \frac{5}{3}u_2, \|\mathrm{x}\| = \sqrt{3}$.

Question 3. Let S be the subspace of $\mathbb{R}^3$ spanned by the vectors $\mathbf{u}_2$ and $\mathbf{u}_3$ of Question 2. Let $\mathbf{x} = (1, 2, 2)^T$. Find the projection $\mathbf{p}$ of $\mathbf{x}$ onto S. Show that $(\mathbf{p} - \mathbf{x}) \perp \mathbf{u}_2$ and $(\mathbf{p} - \mathbf{x}) \perp \mathbf{u}_3$.

Solution

Projection p of $\mathrm{x} = (1, 2, 2)^T$ onto S is given by $\mathrm{p} = UU^T\mathbf{x}$ where

$$U = \begin{bmatrix} u_1 & u_2 \end{bmatrix} = \begin{bmatrix} 2/3 & 1/\sqrt{2} \\ 2/3 & -1/\sqrt{2} \\ 1/3 & 0 \end{bmatrix} \begin{bmatrix} 2/3 & 2/3 & 1/3 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} =$$

$$\begin{bmatrix} \frac{4}{9} + \frac{1}{2} & \frac{4}{9} - \frac{1}{2} & \frac{2}{9} + 0 \\ \frac{4}{9} - \frac{1}{2} & \frac{4}{9} + \frac{1}{2} & \frac{2}{9} + 0 \\ \frac{2}{9} + 0 & \frac{2}{9} - 0 & \frac{1}{9} + 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{17}{18} & -\frac{1}{18} & \frac{2}{9} \\ -\frac{1}{18} & \frac{17}{18} & \frac{2}{9} \\ \frac{2}{9} & \frac{2}{9} & \frac{1}{9} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \frac{1}{18} \begin{bmatrix} 23 \\ 41 \\ 16 \end{bmatrix}$$

As $\mathrm{p} = \frac{1}{18} \begin{bmatrix} 23 \\ 41 \\ 16 \end{bmatrix}$ and $\mathbf{x} = (1, 2, 2)^T$, get

$$\mathrm{p} - \mathrm{x} = \frac{1}{18} \begin{bmatrix} 23 \\ 41 \\ 16 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$= \frac{1}{18} \begin{bmatrix} 5 \\ 5 \\ -20 \end{bmatrix}$$

To show that $(\mathbf{p} - \mathrm{x}) \perp \mathbf{u}_2$, consider $(\mathbf{p} - \mathrm{x}) \cdot \mathbf{u}_2$

$$\begin{aligned} (\mathbf{p} - \mathbf{x}) \cdot \mathbf{u}_2 &= \frac{1}{18} \begin{bmatrix} 5 \\ 5 \\ -20 \end{bmatrix} \begin{bmatrix} 2/3 \\ 2/3 \\ 1/3 \end{bmatrix} \\ &= \frac{1}{18} \left(\frac{10}{3} + \frac{10}{3} - \frac{20}{3} \right) \\ &= 0 \end{aligned}$$

From the above calculation, we observe that $(\mathbf{p} - \mathbf{x}) \cdot \mathbf{u}_2 = 0$ Therefore, $(\mathrm{p} - \mathrm{x}) \perp \mathrm{u}_2$

Question 4. Let $\mathbf{u}_1$ and $\mathbf{u}_2$ form an orthonormal basis for $\mathbb{R}^2$ and let $\mathbf{u}$ be a unit vector in $\mathbb{R}^2$. If $\mathbf{u}^T\mathbf{u}_1 = \frac{1}{2}$, determine the value of $\left|\mathbf{u}^T\mathbf{u}_2\right|$.

Solution

Let u_1 and u_2 form an orthonormal basis for R^2 and let u be a unit vector in R^2

If $c_1 = u^T u_1 = \frac{1}{2}$ and $c_2 = u^T u_2$ then by the theorem let $\{u_1, u_2 \dots u_n\}$ be an orthonormal basis for an inner product space v. If $v = \sum_{i=1}^{n} c_i u_i$ then $c_i = \langle v, u_i \rangle$ We have $u = c_1 u_1 + c_2 u_2$. It follows from parseval's formula that

$$\begin{aligned} 1 &= \|u\|^2 \\ &= c_1^2 + c_2^2 \\ &= \frac{1}{4} + c_2^2 \\ \Rightarrow \frac{3}{4} &= c_2^2 \end{aligned}$$

Hence $\left|u^T u_2\right| = |c_2| = \frac{\sqrt{3}}{2}$

Question 5. Let $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be an orthonormal basis for an inner product space V. If $\mathbf{x} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3$ is a vector with the properties $\|\mathbf{x}\| = 5, \langle \mathbf{u}_1, \mathbf{x} \rangle = 4$, and $\mathbf{x} \perp \mathbf{u}_2$, then what are the possible values of c_1, c_2, c_3 ?

Solution

By Perceval's formula:

$$c_1^2 + c_2^2 + c_3^2 = \|x\|^2 = 5^2 = 25$$

Consider the following theorem: "Let $\{u_1, u_2, \dots u_n\}$ be an orthonormal basis for an inner product space V. If $v = \sum_{i=1}^{n} c_i u_i$ then $c_i = \langle v_i u_i \rangle^m$ Use above theorem to find the value of constants as follows:

$$c_1 = \langle u_1, x \rangle = 4$$

And

$$c_2 = \langle u_2, x \rangle = 0$$

As $c_1^2 + c_2^2 + c_3^2 = 25$, therefore

$$\begin{aligned} 4^2 + 0^2 + c_3^2 &= 25 \\ c_3^2 &= 9 \\ c_3 &= \pm 3 \end{aligned}$$

Therefore, the values are $c_1 = 4, c_2 = 0, c_3 = \pm 3$

Question 6. For each of the following, use the Gram-Schmidt process to find an orthonormal basis for Row(A).

(a) $A = \begin{pmatrix} -1 & 3 \\ 1 & 5 \end{pmatrix}$

(b) $A = \begin{pmatrix} 2 & 5 \\ 1 & 10 \end{pmatrix}$

Solution

(a)

$$\left\{ \left(-\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \right)^T, \left(\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right)^T \right\}$$

(b)

$$\left\{ \left(\frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} \right)^T, \left(\frac{-5}{\sqrt{29}}, \frac{2}{\sqrt{29}} \right)^T \right\}$$

Question 7. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 12 \\ 6 \\ 18 \end{pmatrix}$

(a) Use the Gram-Schmidt process to find an orthonormal basis for the column space of A.

(b) Factor A into a product QR, where Q has an orthonormal set of column vectors and R is upper triangular.

(c) Solve the least squares problem $A\mathrm{x} = \mathrm{b}$

Solution

(a) The orthonormal basis is

$$\left\{ \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \right)^T, \left(\frac{-\sqrt{2}}{6}, \frac{4\sqrt{2}}{6}, \frac{-\sqrt{2}}{6} \right)^T \right\}$$

$$= \left\{ \frac{1}{3}(2, 1, 2)^T, \frac{\sqrt{2}}{6}(-1, 4, -1)^T \right\}$$

(b) We can factorize A into the product θR, where

$$R = \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix} = \begin{bmatrix} 3 & 5/3 \\ 0 & \sqrt{2}/3 \end{bmatrix}$$

$$\theta = \begin{bmatrix} 2/3 & -\sqrt{2}/6 \\ 1/3 & 2\sqrt{2}/3 \\ 2/3 & -\sqrt{2}/6 \end{bmatrix}$$

c) $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}, b = \begin{bmatrix} 12 \\ 6 \\ 18 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 12 \\ 6 \\ 18 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} 9 & 5 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 66 \\ 36 \end{bmatrix}$$

$$9x_1 + 5x_2 = 66$$

$$5x_1 + 3x_2 = 36$$

Multiply first equation with 5 and second equation with 9 and then subtract, we get $x_1 = 9, x_2 = -3$. Therefore $x = \begin{bmatrix} 9 \\ -3 \end{bmatrix}$

Question 8. Find the eigenvalues and the corresponding eigenspaces for each of the following matrices: (a) $\begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 6 & -4 \\ 3 & -1 \end{pmatrix}$

(c) $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

(e) $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & 5 & -1 \end{pmatrix}$

Solution

Solution For an $n \times n$ matrix A, its eigenvalues are the solutions to the characteristic equation $\det(A - \lambda I) = 0$, and the corresponding eigenspace for eigenvalue λ is the null space of $A - \lambda I$.

(a) Eigenvalues are $\lambda_1 = 5, \lambda_2 = -1$.

Corresponding eigenspaces are Span $\left\{(1,1)^T\right\}$ and Span $\left\{(1,-2)^T\right\}$.

(b) Eigenvalues are $\lambda_1 = 2, \lambda_2 = 3$.

Corresponding eigenspaces are Span $\left\{(1,1)^T\right\}$ and Span $\left\{(4,3)^T\right\}$.

(c) Eigenvalue is $\lambda = 0$.

Corresponding eigenspace is Span $\left\{(1,0,0)^T\right\}$.

(d) Eigenvalues are $\lambda_1 = 2, \lambda_2 = 1$.

Corresponding eigenspaces are: Span $\left\{(1,1,0)^T\right\}$ and Span $\left\{(1,0,0)^T, (0,1,-1)^T\right\}$

(e) Eigenvalues are $\lambda_1 = -2, \lambda_2 = 1$ and $\lambda_3 = 4$

Corresponding eigenspaces are Span $\left\{(1,1,-5)^T\right\}$, Span $\left\{(1,0,0)^T\right\}$ and Span $\left\{(1,1,1)^T\right\}$

Question 9. Let A be an $n \times n$ matrix. Prove that A is singular if and only if $\lambda = 0$ is an eigenvalue of A.

Solution

Proof. A is singular if and only if the homogenous equation $A\mathrm{x} = 0$ has a non-zero solution. If such a solution exist, and let it be x_0, then we have $A\mathrm{x}_0 = 0\mathrm{x}_0$ and $\lambda = 0$ is an eigenvalue of A. Conversely, an eigenvalue 0 implies the equation $A\mathrm{x} = 0\mathrm{x}$ has a non-zero solution, which indicates A is singular.

Question 10. Let A be a nonsingular matrix and let λ be an eigenvalue of A. Show that $1/\lambda$ is an eigenvalue of A^{-1}.

Solution

Proof. λ is an eigenvalue of A means the equation $A\mathrm{x} = \lambda\mathrm{x}$ has non-zero solution for x. Let x_0 be such a solution (an eigenvector of A), let $\mathrm{y}_0 =$

$A\mathrm{x}_0 = \lambda \mathrm{x}_0$. Because A is invertible, we have $\mathrm{y}_0 \neq 0$ and $A^{-1}\mathbf{y}_0 = A^{-1}A\mathrm{x}_0 = \mathrm{x}_0 = \frac{1}{\lambda}\mathrm{y}_0$, therefore, $1/\lambda$ is an eigenvalue of A^{-1}.

Question 11. An $n \times n$ matrix A is said to be idempotent if $A^2 = A$. Show that if λ is an eigenvalue of an idempotent matrix, then λ must be either 0 or 1 .

Solution

Eigenvalues are solutions to the equation $|A-\lambda I| = 0$. For idempotent matrix A, we can construct the following multiplication and have:

$$(A - \lambda I)(A + (\lambda - 1)I) = A^2 - A - \lambda(\lambda - 1)I = -\lambda(\lambda - 1)I$$

Then, we have the determinant $|(A - \lambda I)(A + (\lambda - 1)I)| = -\lambda(\lambda - 1)$. Now, consider if λ is an eigenvalue of A, we must have $|A - \lambda I| = 0$ and because

$$-\lambda(\lambda - 1) = |(A - \lambda I)(A + (\lambda - 1)I)| = |(A - \lambda I)||(A + (\lambda - 1)I)| = 0,$$

λ must be either 0 or 1 .

Question 12. Let A be an $n \times n$ matrix and let $B = A + I$. Is it possible for A and B to be similar? Explain.

Solution

Because $B = A + I$, the characteristic equation for A is $\det(A - \lambda I) = 0$ and for B is $\det(B-\lambda I) = \det(A-(\lambda-1)I) = 0$. We see that if λ_0 is an eigenvalue of A, then $\lambda_0 + 1$ is an eigenvalue of B. Since A and B have different sets of eigenvalues, they cannot be similar.

Question 13. Let Q be an orthogonal matrix.

(a) Show that if λ is an eigenvalue of Q, then $|\lambda| = 1$.

(b) Show that $|\det(Q)| = 1$.

Solution

(a) Proof. If λ is an eigenvalue of Q, and let x_0 be a corresponding eigenvector, then we have $Q\mathbf{x}_0 = \lambda \mathbf{x}_0$ and $\mathbf{x}_0 \neq \mathbf{0}$. Because Q is an orthogonal matrix, we

have:

$$\lambda^2 \mathbf{x}_0^T \mathbf{x}_0 = (\lambda \mathbf{x}_0)^T (\lambda \mathbf{x}_0) = (Q\mathbf{x}_0)^T (Q\mathbf{x}_0) = \mathbf{x}_0^T \left(Q^T Q\right) \mathbf{x}_0 = \mathbf{x}_0^T \mathbf{x}_0$$

Because $\mathbf{x}_0^T \mathbf{x}_0 \neq 0$, the above equation implies $\lambda^2 = 1$, or $|\lambda| = 1$.
(b) Proof. For orthogonal matrix Q, we have $Q^T = Q^{-1}$, and,
$(\det(Q))^2 = \det(Q) \det\left(Q^T\right) = \det(Q) \det\left(Q^{-1}\right) = \det\left(QQ^{-1}\right) = \det(I) = 1$.
Hence, $|\det(Q)| = 1$.

Question 14. Prove that, for any $\mathbf{u}$ and $\mathbf{v}$ in an inner product space V,

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$$

Give a geometric interpretation of this result for the vector space $\mathbb{R}^2$.

Solution

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$$
$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$$
$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2 = 2\left(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\right)$$

If the vectors $\mathbf{u}$ and $\mathbf{v}$ are used to form a parallelogram in the plane, then the diagonals will be $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$. The equation shows that the sum of the squares of the lengths of the diagonals is twice the sum of the squares of the lengths of the two sides.

Question 15. Consider the vector space $\mathbb{R}^n$ with inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}$. Show that for any $n \times n$ matrix A,
(a) $\langle A\mathbf{x}, \mathbf{y} \rangle = \left\langle \mathbf{x}, A^T \mathbf{y} \right\rangle$
(b) $\left\langle A^T A\mathbf{x}, \mathbf{x} \right\rangle = \|A\mathbf{x}\|^2$

Solution

(a) $\langle A\mathbf{x}, \mathbf{y} \rangle = (A\mathbf{x})^T \mathbf{y} = \mathbf{x}^T A^T \mathbf{y} = \left\langle \mathbf{x}, A^T \mathbf{y} \right\rangle$
(b) $\left\langle A^T A\mathbf{x}, \mathbf{x} \right\rangle = \left\langle \mathbf{x}, A^T A\mathbf{x} \right\rangle = \mathbf{x}^T A^T A\mathbf{x} = (A\mathbf{x})^T A\mathbf{x} = \langle A\mathbf{x}, A\mathbf{x} \rangle = \|A\mathbf{x}\|^2$

Question 16. Let $\mathbf{u}$ be a unit vector in $\mathbb{R}^n$ and let $H = I - 2\mathbf{u}\mathbf{u}^T$. Show that H is both orthogonal and symmetric and hence is its own inverse.

Solution

$$
\begin{aligned}
H^T &= \left(I - 2\mathbf{u}\mathbf{u}^T\right)^T = I^T - 2\left(\mathbf{u}^T\right)^T \mathbf{u}^T = I - 2\mathbf{u}\mathbf{u}^T = H \\
H^T H &= H^2 \\
&= \left(I - 2\mathbf{u}\mathbf{u}^T\right)^2 \\
&= I - 4\mathbf{u}\mathbf{u}^T + 4\mathbf{u}\mathbf{u}^T\mathbf{u}\mathbf{u}^T \\
&= I - 4\mathbf{u}\mathbf{u}^T + 4\mathbf{u}\mathbf{u}^T \\
&= I
\end{aligned}
$$

Question 17. Given $\mathbf{x}_1 = \frac{1}{2}(1,1,1,-1)^T$ and $\mathbf{x}_2 = \frac{1}{6}(1,1,3,5)^T$, verify that these vectors form an orthonormal set in $\mathbb{R}^4$. Extend this set to an orthonormal basis for $\mathbb{R}^4$ by finding an orthonormal basis for the null space of

$$\begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 3 & 5 \end{pmatrix}$$

[Hint: First find a basis for the null space and then use the Gram-Schmidt process.]

Solution

First, we find the nullspace of the matrix by first row reducing the matrix.

$$\begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 3 & 5 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

Thus,

$$\text{nullspace} \begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 3 & 5 \end{pmatrix} = \text{Span}\left(\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 0 \\ -3 \\ 1 \end{pmatrix}\right)$$

Now, we apply the Gram-Schmidt process to these two vectors. Let

$$\mathbf{x}_3 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } \mathbf{x}_4 = \begin{pmatrix} 4 \\ 0 \\ -3 \\ 1 \end{pmatrix}$$

$$\mathbf{u}_3 = \frac{\mathbf{x}_3}{\|\mathbf{x}_3\|} = \frac{\mathbf{x}_3}{\sqrt{1+1}} = \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \end{pmatrix}$$

The projection of $\mathbf{x}_4$ onto $\mathbf{u}_3$ is

$$\mathbf{p}_3 = \langle \mathbf{x}_4, \mathbf{u}_3 \rangle \, \mathbf{u}_3 = -\frac{4}{\sqrt{2}}\mathbf{u}_3 = \begin{pmatrix} 2 \\ -2 \\ 0 \\ 0 \end{pmatrix}$$

Then, $\mathbf{x}_4 - \mathbf{p}_3$ is orthogonal to $\mathbf{u}_3$.

$$\mathbf{x}_4 - \mathbf{p}_3 = \begin{pmatrix} 2 \\ 2 \\ -3 \\ 1 \end{pmatrix}$$

Then,

$$\mathbf{u}_4 = \frac{\mathbf{x}_4 - \mathbf{p}_3}{\|\mathbf{x}_4 - \mathbf{p}_3\|} = \frac{\mathbf{x}_4 - \mathbf{p}_3}{\sqrt{4+4+9+1}} = \begin{pmatrix} \frac{2}{\sqrt{18}} \\ \frac{2}{\sqrt{18}} \\ \frac{-3}{\sqrt{18}} \\ \frac{1}{\sqrt{18}} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{3} \\ \frac{\sqrt{2}}{3} \\ -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{6} \end{pmatrix}$$

Thus, an orthonormal basis for the nullspace of the matrix is

$$\left\{ \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{\sqrt{2}}{3} \\ \frac{\sqrt{2}}{3} \\ -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{6} \end{pmatrix} \right\}$$

Thus, an orthonormal basis for R^4 is

$$\left\{ \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{-1}{2} \end{pmatrix}, \begin{pmatrix} \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{2} \\ \frac{5}{6} \end{pmatrix}, \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{\sqrt{2}}{3} \\ \frac{\sqrt{2}}{3} \\ -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{6} \end{pmatrix} \right\}$$

Question 18. Show that if two $n \times n$ matrices A and B have a common eigenvector $\mathbf{x}$ (but not necessarily a common eigenvalue), then $\mathbf{x}$ will also be an eigenvector of any matrix of the form $C = \alpha A + \beta B$.

Solution

If $\mathbf{x}$ is an eigenvector of A belonging to the eigenvalue λ and $\mathbf{x}$ is also an eigenvector of B corresponding to the eigenvalue μ, then

$$(\alpha A + \beta B)\mathbf{x} = \alpha A\mathbf{x} + \beta B\mathbf{x} = \alpha\lambda\mathbf{x} + \beta\mu\mathbf{x} = (\alpha\lambda + \beta\mu)\mathbf{x}$$

Therefore $\mathbf{x}$ is an eigenvector of $\alpha A + \beta B$ belonging to $\alpha\lambda + \beta\mu$.